Coding Witness Fibers

Exact reductions, shared-evaluation tricks, and barriers for existential circuit projection

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Abstract

Let $N \geq 1$, $M \geq 0$, let $f : \{0,1\}^N \times \{0,1\}^M \to \{0,1\}$ be a Boolean verifier, and let $g(x) = \bigvee_y f(x,y)$. We ask when the 2^M witness branches can be compressed into a small deterministic circuit for g. A companion mass-production theorem can batch all branches whenever the deterministic input occupies a fixed positive fraction of the verifier input, but its resulting bound is worse by a factor 2^M than direct synthesis of the N-input function g. The useful change of viewpoint is therefore to code the entire witness fiber

$$v_x = (f(x,y))_{y \in \{0,1\}^M},$$

not merely to batch its coordinates.

This viewpoint gives three exact reductions. If the span of the realized fibers has dimension R, an information set of R actual witnesses determines whether every fiber is zero. If every nonempty fiber has density at least δ and there are K distinct nonempty fibers, a fixed hitting set of about $(\log(K+1))/\delta$ actual witnesses suffices. Without either promise, an f-dependent linear map with at most $\min\{N, 2^M\}$ output bits separates every realized nonzero fiber from zero; this worst-case measurement count is exact. With one-sided distributional error $0 < \rho \leq 1$, only $\lceil \log_2(1/\rho) \rceil$ random parity measurements are needed.

The last reduction compresses measurements, not gates. A recovery-boundary identity does expose a shared-input trick: when parity branches are decoded through a linear local-recovery code, repeated resource incidences cancel by symmetric difference. Its circuit benefit is conditional on the surviving resource functions having a cheap joint evaluator. We then prove an exact obstruction: if one fixed linear sketch supports a correct update through coordinatewise AND on all profiles, its kernel is a coordinate ideal. Up to a change of basis, such a sketch merely retains actual witness coordinates. Thus a stronger construction must exploit reachable profiles, change encodings between gates, tolerate error, or use a projection-stable representation. We derive concrete bounds for dense fibers, low witness degree, and supplied decomposable or bounded-width knowledge-compilation representations, and isolate the remaining aggregate-synthesis problem. The elementary coding arguments are not claimed as literature-novel; the purpose of this note is to make the reductions, barrier, cancellation opportunity, and open target precise.

1 The question and the answer in one page

Fix a circuit for

$$f : \{0,1\}^N \times \{0,1\}^M \longrightarrow \{0,1\}, \qquad g(x) = \bigvee_{y\in\{0,1\}^M} f(x,y), \tag{1.1}$$

and write $S = \mathrm{C}(f)$. Throughout, $N \geq 1$ and $M \geq 0$; if $N = 0$, the projection is one free constant and needs no circuit. The x bits are the deterministic input and the y bits are nondeterministic choices. The elementary simulation uses one restricted copy of f per witness:

$$\mathrm{C}(g) \leq 2^M S + 2^M - 1. \tag{1.2}$$

The central question is whether common input and coding let us replace the factor 2^M by something smaller.

The answer has two layers. At the level of information, yes: the 2^M witness branches can often be replaced by a short list of actual witnesses, and they can always be replaced by at most N parity measurements when $N \leq 2^M$. At the level of unrestricted circuit gates, not automatically: forming a parity such as

$$x \longmapsto \bigoplus_{y\in T} f(x,y) \tag{1.3}$$

may itself contain essentially all of the work.

Here is the map. Put $D = 2^M$; let K count the distinct nonempty witness fibers, let R be their span dimension, and let δ be their minimum relative weight; $0 < \rho \leq 1$ is a target distributional error. If $K = 0$, then $g = 0$ is free, so the nontrivial rows below assume $K \geq 1$.

Method	Hypothesis and dependence	Summaries	Gate conclusion
Full enumeration	none	D actual witnesses	$D(S+1)-1$ gates
Span information set	f-dependent; fiber-span dimension R	R actual witnesses	$R(S+1)-1$ gates
Dense-fiber hitting	density δ; f-dependent	$O((\log(K+1))/\delta)$ actual witnesses	same factor times $S+1$
Exact linear fingerprint	no promise; f-dependent masks	$O(\log(K+1))$ parities; at most $\min\{N,D\}$	exact zero test; aggregate cost unpaid
Approximate fingerprint	fixed μ: f,μ-dependent mask; or a fresh mask per evaluation	$\lceil\log_2(1/\rho)\rceil$ parities	one-sided error ρ; aggregate cost unpaid
DNNF forgetting	a decomposable NNF (DNNF) is supplied	decomposable subcircuits	linear-size projection from a supplied DNNF

The coding move. Regard $x \mapsto v_x$ as a codebook of at most 2^N realized words, and choose an f-dependent linear map whose kernel avoids every realized nonzero word. This replaces the D branch values by $O(\log(K+1))$ parity aggregates. It is an exact compression of the zero test, not yet a smaller circuit: the aggregates still have to be computed from the succinct verifier. Theorem 4.1 proves the compression. In Proposition 4.4, local-recovery equations cancel resource coordinates used an even number of times, leaving a symmetric-difference boundary whose joint evaluator is still charged. Finally, Theorem 5.1 rules out universal propagation by one fixed linear sketch.

The most useful conceptual distinction is

$$\boxed{\text{few summaries} \quad \neq \quad \text{few gates to produce them.}} \tag{1.4}$$

The ideal-kernel theorem in Section 5 makes this distinction formal for fixed linear sketches and AND gates.

Contents

Scope, contribution, and research process

Contribution and scope. The contribution claimed here is the integrated reduction-and-barrier framework, including the recovery-boundary bridge from local recovery to common-input parity aggregation, and the resulting specification of the aggregate-synthesis target. No individual elementary coding lemma is claimed as literature-novel, and an unsuccessful search is not used as evidence of priority.

Status of the claims. Formal statements are marked PROVED HERE, IMPORTED, or KNOWN. External facts and comparisons in the prose are cited where they are used. Conditional reductions and unresolved directions are called conditional or open in the surrounding text. The only imported technical theorem is the mass-production estimate from the companion manuscript [20].

Research-process disclosure. This investigation was developed with substantive AI assistance. The underlying mass-production project began in a ChatGPT.com conversation that the author exported as a LaTeX document. Later interactive sessions used Claude Fable 5 and 5.1 through Claude Code and GPT-5.6 Sol through Codex for experiments, broad literature searches, proof ideas, counterexample searches, and drafting. The author checked the arguments and citations and is responsible for the note.

2 Baselines and what mass production does not yet buy

We use fan-in-two $\{\wedge, \vee, \neg\}$ circuits, with free constants, fan-out, and output designations; size counts gates. For a Boolean map h with any finite output length, $\mathrm{C}(h)$ is its minimum size. Let

$$L(N) = \max_{h:\{0,1\}^N \to \{0,1\}} \mathrm{C}(h).$$

In this convention, as $N \to \infty$, classical synthesis gives

$$L(N) \le (1 + o(1))\frac{2^N}{N}; \tag{2.1}$$

the sharp standard-basis statement used here is recorded by Frandsen and Miltersen [9], following Lupanov's synthesis method [14].

For $t \ge 1$, let $f^{\times t}$ evaluate f on t independent input blocks, and set $\mathrm{MP}_t(f) = \mathrm{C}(f^{\times t})$.

Proposition 2.1 (The unconditional envelope; PROVED HERE)**.** *For every N, M and f as in* (1.1),

$$\mathrm{C}(g) \le \min\left\{2^M \mathrm{C}(f) + 2^M - 1, L(N), \mathrm{MP}_{2^M}(f) + 2^M - 1\right\}. \tag{2.2}$$

Proof. For the first bound, hardwire a different witness into each of 2^M copies of a circuit for f, identify all copies of the x inputs, and join the outputs with a binary OR tree. For the second, synthesize the N-input function g directly. For the third, begin with a circuit for $f^{\times 2^M}$, identify its deterministic input blocks, hardwire its witness blocks to all distinct values, and append the OR tree. Restrictions, input identification, and repeated output designations add no gates. □

2.1 The imported mass-production estimate

The companion manuscript proves the following independent-input statement.

Theorem 2.2 (Exponential-range mass production; IMPORTED)**.** *For every fixed $0 \le \gamma < 1$, there is A_γ such that every $h : \{0,1\}^n \to \{0,1\}$ and every $1 \le t \le 2^{\gamma n}$ satisfy*

$$\mathrm{MP}_t(h) \le A_\gamma \frac{2^n}{n}. \tag{2.3}$$

The high-rate version has normalized leading coefficient at most $1/(1-\gamma)$ [20].

Apply this with $n = N + M$ and $t = 2^M$. One fixed $\gamma < 1$ suffices along a family precisely when

$$\frac{M}{N+M} \le \gamma, \tag{2.4}$$

so the deterministic fraction remains bounded away from zero. Along a sequence with $N + M \to \infty$ and $M/(N+M) \to \theta < 1$, first use any fixed $\gamma > \theta$ and then let $\gamma \downarrow \theta$. Since $(1-\theta)(N+M) = N + o(N+M)$, the imported coefficient gives

$$\mathrm{C}(g) \le (1 + o(1)) \frac{2^{N+M}}{N}. \tag{2.5}$$

The appended OR tree is lower order because $2^M/(2^{N+M}/N) = N/2^N = o(1)$. But direct synthesis gives $(1 + o(1))2^N/N$. The mass-production route is therefore worse by the factor $(1 + o(1))2^M$. This compares upper bounds, not the true complexity of a particular g.

The conclusion is useful because it prevents a false start: the companion theorem does batch all witnesses in the fixed-ratio regime, but that fact alone gives no new worst-case determinization bound. Existential projection has reduced the exposed input length from $N + M$ to N, and generic synthesis already exploits that reduction.

2.2 What identical deterministic inputs do remove

The common x is not irrelevant. It removes repeated evaluations of the same resource function at the same input.

Lemma 2.3 (Identical-call collapse; PROVED HERE)**.** *Suppose a multi-output computation requests functions $H_{i_1}, \dots, H_{i_s} : \{0,1\}^N \to \{0,1\}$ all on the same input x. Replacing the list by one representative of every distinct extensional function changes neither minimum circuit size nor the obtainable outputs, apart from free output designations.*

Proof. From representatives to the original list, repeat each representative output wire wherever it is requested. In the other direction, discard duplicate outputs. Both operations are free in the circuit model. □

For existential projection, every branch supplies the same x. Duplicate requests for the same restricted function of x can therefore be coalesced by free fan-out, but up to 2^M distinct restricted functions may remain. Common input compresses *call multiplicity*; it does not by itself compress *function multiplicity*. The stronger parity-specific cancellation available inside a linear recovery code is isolated in Section 4.2.

3 The fiber code: dimension and distance from zero

Put $D = 2^M$ and index the coordinates of $\mathbb{F}_2^D$ by witnesses. For each $x \in \{0,1\}^N$, define its witness fiber vector

$$v_x = (f(x,y))_{y \in \{0,1\}^M} \in \mathbb{F}_2^D. \tag{3.1}$$

Then $g(x) = 1$ if and only if $v_x \neq 0$. Let

$$\mathcal{V}_f = \{v_x : v_x \neq 0\}, \qquad K = |\mathcal{V}_f|, \qquad U_f = \operatorname{span}(\mathcal{V}_f), \qquad R = \dim U_f. \tag{3.2}$$

Thus U_f is a binary linear code of length D. The realized nonzero fibers need not include all of its nonzero codewords.

3.1 Dimension gives actual witnesses

Lemma 3.1 (Information set; PROVED HERE)**.** *If $U \leq \mathbb{F}_2^D$ has dimension R, there is a coordinate set $H \subseteq [D]$ of size R such that restriction $U \to \mathbb{F}_2^H$ is injective.*

Proof. Put a basis of U in the rows of an $R \times D$ matrix. The matrix has R linearly independent columns. Their indices form H, and restriction to those columns maps the chosen basis to an invertible $R \times R$ matrix. □

Translating coordinate indices back to witnesses gives an immediate circuit bound.

Corollary 3.2 (Span-dimension determinization; PROVED HERE)**.** *If $R > 0$, there is a fixed set $H \subseteq \{0,1\}^M$ of R witnesses such that*

$$g(x) = \bigvee_{y \in H} f(x,y), \qquad \mathrm{C}(g) \leq R\,\mathrm{C}(f) + R - 1. \tag{3.3}$$

If $R = 0$, then g is constant zero.

This is a genuine gate bound: the summaries are actual restricted copies of the verifier. It is nonuniform because the information set may depend on the complete relation. Given the explicit fibers, Gaussian elimination finds it; no efficient procedure from a succinct circuit for f is implied.

3.2 Realized-fiber density gives a hitting set

For a realized fiber, write $F_x = \{y : f(x,y) = 1\}$. Suppose every nonempty F_x has relative size at least δ. The number of distinct nonempty sets is K, not necessarily 2^N.

Theorem 3.3 (Dense-fiber hitting; PROVED HERE)**.** *If $K \geq 1$ and $0 < \delta < 1$, there is a fixed witness set H with*

$$|H| \leq \min\left\{D, \left\lfloor \frac{\ln K}{-\ln(1-\delta)} \right\rfloor + 1\right\} \leq \min\left\{D, \frac{\ln K}{\delta} + 1\right\}. \tag{3.4}$$

It intersects every nonempty fiber. For $\delta = 1$, one witness suffices. Consequently,

$$\mathrm{C}(g) \leq |H|(\mathrm{C}(f) + 1) - 1. \tag{3.5}$$

Proof. Choose h uniform independent witnesses. A fixed fiber is missed with probability at most $(1-\delta)^h$. A union bound over the K distinct fibers makes the failure probability less than one for the displayed value of h. Remove repetitions, or use all D witnesses if that is shorter. The second inequality uses $-\ln(1-\delta) \geq \delta$. Finally, hardwire the witnesses in H into $|H|$ verifier copies and OR their outputs. $\square$

The same proof has a greedy interpretation: among the fibers not yet hit, some witness lies in at least a δ fraction. This constructs H from an explicit relation, but the required coverage queries may themselves be exponential from a succinct verifier.

If the set system $\{F_x\}$ has VC dimension v, put $\bar{v} = \max\{1, v\}$. The epsilon-net theorem of Haussler and Welzl [11], with its endpoint cases absorbed explicitly, gives the boundary-safe alternative size

$$O\left(\frac{\bar{v}}{\delta}\left(1 + \log\frac{1}{\delta}\right)\right) \tag{3.6}$$

under the uniform measure on the witness cube. The measure and the common density promise are essential. This refinement is useful only when its expression beats both (1.2) and direct synthesis.

Dimension and distance are complementary. An information set works for every nonzero vector in the entire span U_f and costs R coordinates. The density argument needs to hit only the K realized fibers and may cost much less than R; it uses their actual weights, which can exceed the minimum distance of U_f.

3.3 Low witness degree

Low degree is a canonical concrete promise under which the density route becomes effective. Every Boolean function of y has a unique multilinear polynomial over $\mathbb{F}_2$. Suppose that, for every x, the polynomial $y \mapsto f(x, y)$ has degree at most an integer $0 \leq d \leq M$. Its evaluation vector lies in the binary Reed–Muller code $\mathrm{RM}(d, M)$ [16, 19].

Lemma 3.4 (Reed–Muller weight; PROVED HERE). *A nonzero multilinear polynomial in M variables of degree at most d is one on at least 2^{M-d} points.*

Proof. The case $M = 0$ is the nonzero constant polynomial of weight one. For $M > 0$, induct on M. Write $p(y_1, \ldots, y_M) = a(y_1, \ldots, y_{M-1}) + y_M b(y_1, \ldots, y_{M-1})$. If $b = 0$, the weight doubles and induction applies to a. If $b \neq 0$, then $\deg b \leq d - 1$. Whenever $b(z) = 1$, exactly one of $p(z, 0)$ and $p(z, 1)$ is one. Induction gives at least $2^{(M-1)-(d-1)} = 2^{M-d}$ such z. $\square$

Thus $\delta \geq 2^{-d}$ in Theorem 3.3, giving an f-dependent witness set of size $O(2^d(\log K + 1))$. There is also an explicit universal information set:

$$H_{M,d} = \{y \in \{0,1\}^M : \mathrm{wt}(y) \leq d\}, \qquad |H_{M,d}| = B(M, d) := \sum_{i=0}^{d} \binom{M}{i}. \tag{3.7}$$

Lemma 3.5 (Low-weight information set; PROVED HERE). *If a degree-at-most-d multilinear polynomial vanishes on $H_{M,d}$, it is zero. Moreover, no smaller coordinate set works for the entire degree class.*

Proof. Write $p(y) = \sum_{|A| \le d} c_A \prod_{i \in A} y_i$. At the indicator 1_S, where $|S| \le d$,

$$p(1_S) = \sum_{A \subseteq S} c_A.$$

Induction on $|S|$ recovers every c_S from these values. Hence restriction to $H_{M,d}$ is injective. The polynomial space has dimension $B(M, d)$, so rank–nullity forbids an injective restriction to fewer coordinates. □

If $K = 0$, then g is constant zero and $\mathrm{C}(g) = 0$. Assume $K \ge 1$ and define the density term

$$Q_d(K) = \begin{cases} 1, & d = 0, \\ \left\lfloor \dfrac{\ln K}{-\ln(1 - 2^{-d})} \right\rfloor + 1, & 1 \le d \le M. \end{cases}$$

Combining the three actual-witness routes gives the exact-form bound

$$\mathrm{C}(g) \le L_f(\mathrm{C}(f) + 1) - 1, \qquad L_f = \min\{D, R, B(M, d), Q_d(K)\}, \tag{3.8}$$

whenever the degree promise holds. For fixed d, $B(M, d) = \Theta(M^d)$, while the relation-dependent density branch is $O(2^d \log(K + 1))$.

4 Zero-preserving parity fingerprints

Span dimension and density can both be unfavorable. Measurement compression is nevertheless always possible because only finitely many fibers are realized.

Theorem 4.1 (Exact linear fingerprint; PROVED HERE)**.** *If $K = 0$, the zero map suffices. If $K \ge 1$, put*

$$r = \min\{D, \max\{1, \lceil \log_2 K \rceil\}\}. \tag{4.1}$$

There is an $\mathbb{F}_2$-linear map $A : \mathbb{F}_2^D \to \mathbb{F}_2^r$ such that

$$Av_x = 0 \iff v_x = 0 \qquad \text{for every } x \in \{0, 1\}^N. \tag{4.2}$$

For $N \ge 1$, one always has $r \le \min\{N, D\}$, and this worst-case bound is exact over all relations f.

Proof. For the upper bound, maintain the nonzero fibers not yet separated from zero by the chosen rows. A uniform linear functional vanishes on each fixed nonzero vector with probability 1/2, so some row leaves at most half of the current vectors unseparated. After $r - 1$ rows, at most two remain. A final functional can be one on a single remaining vector. It can also be one on two, because two distinct nonzero vectors over $\mathbb{F}_2$ are linearly independent. This separates the last vectors. The inequality $K \le 2^N$ gives $r \le N$, and $K \le 2^D - 1$ gives $r \le D$.

For tightness, first suppose $D \le N$. A relation can realize every vector of $\mathbb{F}_2^D$ among its fibers, forcing any successful A to be injective and to have at least D output bits. If $D > N$, realize zero and every nonzero vector of an N-dimensional subspace, using exactly 2^N deterministic inputs. The restriction of A to that subspace must be injective, so it needs at least N output bits. □

The construction is deterministic given the explicit fiber list: at each step choose a row attaining the averaging bound. It is not claimed to be efficient from a succinct circuit for f.

The zero-collision argument is a specialization of standard linear-hashing ideas. Universal and linear hashing study broader collision and load goals [3, 1]; succinct perfect linear hashing can even separate every pair in a prescribed sparse set [6]. Here only the pairs $(0, v)$ for $v \in \mathcal{V}_f$ must be separated, and no efficient hash construction from the succinct verifier is asserted.

For the rest of this section assume $K \geq 1$, so $r \geq 1$; the case $K = 0$ has $g = 0$ and needs no circuit.

Write the rows of A as $a_1, \ldots, a_r$ and define

$$h_i(x) = \langle a_i, v_x \rangle = \bigoplus_{y: a_i(y)=1} f(x, y). \tag{4.3}$$

Then

$$g(x) = \bigvee_{i=1}^{r} h_i(x). \tag{4.4}$$

Equivalently, after choosing a basis of $\mathbb{F}_{2^r}$, the columns of A are weights $\alpha_y \in \mathbb{F}_{2^r}$ and

$$g(x) = 1 \quad \Longleftrightarrow \quad \sum_y \alpha_y f(x, y) \neq 0. \tag{4.5}$$

One field symbol still consists of r Boolean wires; the notation does not make the sum cheap.

4.1 Width depends only on the target error

The exact map must work on all x at once. For distributional approximation of g, the width loses its dependence on N.

Theorem 4.2 (Approximate parity fingerprint; PROVED HERE)**.** *Let μ be any fixed distribution on $\{0,1\}^N$ and let $t \geq 0$ be an integer. Choose t independent uniform rows and define*

$$\widetilde{g}_A(x) = 1[Av_x \neq 0].$$

For every A, $\widetilde{g}_A$ has no false positives. Moreover,

$$\mathbb{E}_A \Pr_{x \sim \mu}[g(x) = 1, \widetilde{g}_A(x) = 0] = 2^{-t} \Pr_{x \sim \mu}[g(x) = 1] \leq 2^{-t}. \tag{4.6}$$

Consequently some fixed, f- and μ-dependent A has one-sided error at most 2^{-t}. A freshly random A has the same error bound pointwise for every fixed input.

Proof. Zero always maps to zero. For $v_x \neq 0$, one uniform row has uniform inner product with v_x, so all t rows vanish with probability 2^{-t}. Interchange the finite expectations over A and x. □

Thus, for $0 < \rho \leq 1$, taking $t = \lceil \log_2(1/\rho) \rceil$ gives false-negative mass at most ρ. This is the standard randomized parity sketch of OR, applied only to the realized witness fibers; linear-sketch and parity-query treatments place it in a broader framework [12, 10]. On the full cube $\mathbb{F}_2^D$, an exact deterministic linear sketch for OR must be injective and needs D rows. Theorem 4.1 escapes that lower bound because only at most 2^N fiber vectors are realized.

Small-bias sample spaces can shorten the random seed while making each nonzero parity nearly unbiased [18]. A constant-distance linear code gives another formulation: encode v, sample codeword coordinates, and use the distance to bound the probability that every sample is zero. These ideas reduce randomness or measurement count, not necessarily the Boolean gates used to produce a measurement from an implicit v_x.

The distinction from Valiant–Vazirani isolation is also important [22]. Isolation intersects a witness set with random constraints and seeks a unique surviving *witness*. Here a row reports the parity of a set and may cancel many accepting witnesses. The exact quantifier order is

$$\forall f \; \exists A \; \forall x,$$

not a randomized per-instance isolation claim.

4.2 Recovery-boundary cancellation on a common input

Linear local recovery offers a sharper common-input trick than duplicate-call collapse. Parity aggregates combine recovery equations over $\mathbb{F}_2$, so a resource incidence used twice disappears.

Lemma 4.3 (Symmetric-difference recovery; PROVED HERE)**.** *Let $E : \mathbb{F}_2^D \to \mathbb{F}_2^Q$ be an injective linear encoding, and let $E^* : \mathbb{F}_2^Q \to \mathbb{F}_2^D$ be its adjoint under the standard inner products. For each message coordinate i, choose a recovery vector b_i satisfying $E^* b_i = e_i$, and put $R_i = \operatorname{supp}(b_i)$. Then every $T \subseteq [D]$ satisfies*

$$\bigoplus_{i \in T} v_i = \bigoplus_{z \in \triangle_{i \in T} R_i} (Ev)_z \qquad (v \in \mathbb{F}_2^D), \tag{4.7}$$

where $\triangle$ denotes symmetric difference.

Proof. Injectivity of E makes E^* surjective, so the stated recovery vectors exist. Set $b_T = \sum_{i \in T} b_i$. Then

$$\langle b_T, Ev \rangle = \langle E^* b_T, v \rangle = \left\langle \sum_{i \in T} e_i, v \right\rangle.$$

Over $\mathbb{F}_2$, the support of b_T is precisely the symmetric difference of the supports of the b_i. □

The affine-line recovery used in the companion construction gives a vivid special case [20, Section 4]. In characteristic two, a degree-at-most-$(q-2)$ polynomial on an affine line L of size q satisfies $P(u) = \sum_{z \in L \setminus \{u\}} P(z)$. If every $u \in T \subseteq L$ is recovered through that same line, then

$$\triangle_{u \in T}(L \setminus \{u\}) = \begin{cases} T, & |T| \text{ even}, \\ L \setminus T, & |T| \text{ odd}. \end{cases} \tag{4.8}$$

In particular, the parity of $q-1$ target values collapses to the one remaining code symbol. The identity applies bitwise after expanding field symbols in an $\mathbb{F}_2$ basis.

The next statement records exactly when this cancellation becomes a circuit bound. For $Z \subseteq [Q]$, let

$$G_Z(x) = \left((Ev_x)_z\right)_{z \in Z}, \qquad B_E(Z) = \mathrm{C}(G_Z).$$

Proposition 4.4 (Recovery-boundary reduction; PROVED HERE)**.** *Let the rows $a_1, \ldots, a_r$ of A separate the realized fibers from zero. For each j, choose $b_j \in \mathbb{F}_2^Q$ with $E^* b_j = a_j$, and let $Z = \bigcup_j \operatorname{supp}(b_j)$. Then, in the standard circuit basis,*

$$\mathrm{C}(g) \le B_E(Z) + 4\sum_{j=1}^{r} \max\{0, \operatorname{wt}(b_j) - 1\} + r - 1. \tag{4.9}$$

Proof. Compute the distinct resource functions in G_Z once. The adjoint identity gives

$$\langle a_j, v_x \rangle = \langle b_j, E v_x \rangle = \bigoplus_{z \in \operatorname{supp}(b_j)} G_z(x).$$

A zero-support parity is the free constant zero. A parity of $s \ge 1$ bits uses $s - 1$ XORs, each implementable by at most four $\{\wedge, \vee, \neg\}$ gates. An OR tree on the r resulting fingerprint bits uses $r - 1$ further gates. □

The proposition suggests jointly choosing the fingerprint rows and their recovery representations to minimize a symmetric-difference boundary. It is specific to the shared input: for independent inputs, two incidences with the same resource coordinate generally carry different values and cannot cancel. It is also only a conditional gain. Defining an encoding whose first coordinates are the desired aggregates makes every b_j sparse tautologically; unless those encoded coordinates inherit a cheap evaluator, $B_E(Z)$ contains the original difficulty.

4.3 Why this is not yet a circuit upper bound

Suppose the r functions in (4.3) have a joint circuit of size B. Appending an OR tree proves

$$\mathrm{C}(g) \le B + r - 1. \tag{4.10}$$

This conditional reduction is exact. The missing theorem is a useful bound on B in terms of S, N, M, and r.

Materializing every $f(x, y)$ first costs at most $D\mathrm{C}(f)$ gates, and direct parity trees add $O(rD)$ more. A short seed describing the masks does not remove these gates. Likewise, applying mass production to materialize all fiber coordinates returns to the bound in Section 2. The new question is whether selected aggregates can be computed *without* expanding the full fiber.

5 Why a fixed sketch cannot simply pass through AND

The witness profiles form the algebra

$$\mathcal{R} = \mathbb{F}_2^D$$

with coordinatewise multiplication $(u \odot v)(y) = u(y)v(y)$. For a wire w of the verifier, let

$$V_w(x) = (w(x, y))_{y \in \{0,1\}^M}.$$

At an AND gate $w = u \wedge v$, $V_w(x) = V_u(x) \odot V_v(x)$. We may replace OR by $u \vee v = u \oplus v \oplus (u \wedge v)$ at constant cost. In the resulting $\{\wedge, \oplus, \neg\}$ description, XOR and NOT are linear or affine on profiles, so AND is the nonlinear obstruction to propagating one fixed linear fingerprint. Recall that a subspace $I \subseteq \mathcal{R}$ is an ideal when $u \odot v \in I$ for every $u \in I$ and every $v \in \mathcal{R}$.

Theorem 5.1 (Ideal-kernel classification; PROVED HERE)**.** *Let $A : \mathcal{R} \to \mathbb{F}_2^r$ be linear. The following are equivalent.*

(i) There is an arbitrary function Φ on $A(\mathcal{R})^2$ such that

$$A(u \odot v) = \Phi(Au, Av) \qquad \textit{for all } u, v \in \mathcal{R}. \tag{5.1}$$

(ii) $\ker A$ *is an ideal of the coordinatewise algebra $\mathcal{R}$.*

Every ideal has the form

$$I_T = \operatorname{span}\{e_y : y \in T\} = \{u : \operatorname{supp}(u) \subseteq T\} \tag{5.2}$$

for a unique set of witness coordinates T.

Proof. If (5.1) holds, $k \in \ker A$, and $v \in \mathcal{R}$, then

$$A(k \odot v) = \Phi(0, Av) = A(0 \odot v) = 0,$$

so the kernel is an ideal. Conversely, if $I = \ker A$ is an ideal and $Au = Au'$, $Av = Av'$, then $u' = u + k$ and $v' = v + \ell$ for $k, \ell \in I$. The difference between $u' \odot v'$ and $u \odot v$ is

$$k \odot v + u \odot \ell + k \odot \ell \in I.$$

Hence $A(u \odot v)$ depends only on Au, Av, and Φ is well defined.

For the classification, if $u \in I$ and $u(y) = 1$, then $e_y = e_y \odot u \in I$. Thus I is spanned by exactly the unit vectors that it contains, proving (5.2). □

Corollary 5.2 (Universal propagation equals witness retention; PROVED HERE)**.** *Assume $K \geq 1$. Suppose A satisfies Theorem 5.1 and also separates every $v \in \mathcal{V}_f$ from zero. Put $H = [D] \setminus T$, where* $\ker A = I_T$. *Then H intersects every nonempty witness fiber,* $\operatorname{rank} A = |H|$, *and, up to an injective linear change of output coordinates, A is exactly the projection retaining the bits indexed by H. In particular,*

$$\mathrm{C}(g) \leq |H|(\mathrm{C}(f) + 1) - 1. \tag{5.3}$$

Proof. The quotient $\mathcal{R}/I_T$ is the coordinate space on H. Since A has kernel I_T, its induced map from that quotient to $A(\mathcal{R})$ is an isomorphism, and its rank is $|H|$. If a realized nonzero fiber vanished on H, it would lie in I_T and be killed by A, a contradiction. Thus H hits every nonempty fiber, and the circuit bound follows by restriction and replication. □

For the smallest example, take $D = 2$ and $A(a, b) = a + b$. The pairs

$$(u, v) = ((1, 0), (1, 0)), \qquad (u', v') = ((1, 0), (0, 1))$$

have the same two input sketches, both equal to one, but the product sketches are one and zero. No update rule can distinguish them.

What the theorem does not rule out. Its quantifiers range over every $u, v \in \mathcal{R}$, it uses the same sketch at both inputs and the output, and it demands exactness. A verifier-specific construction may need updates only on profile pairs reached by a common x; it may use a different code at every wire; or it may tolerate error. Those are real escape routes. The theorem rules out only the tempting universal shortcut of pushing one short mixed-parity sketch through arbitrary ANDs.

6 Structural escapes and their limits

6.1 Changing profile spaces: route and dominance check

A gatewise code must pay for its transitions, not just its state names. Suppose each wire w has an encoding $q_w : \{0,1\}^N \to \{0,1\}^{r_w}$ of its realized witness profile, each gate has a transition circuit of size τ_w, and a final circuit of size τ_{out} tests whether the output profile is zero. Composing the transition circuits in the verifier DAG gives

$$\mathrm{C}(g) \le \tau_{\text{out}} + \sum_w \tau_w + \text{leaf-encoding cost.} \tag{6.1}$$

This is bookkeeping, but it states the open target correctly: short profile names, cheap transitions, and an exact zero test are all required.

Output-span dimension by itself will not improve the nonuniform gate bound. If the realized output profiles span a positive-dimensional space of dimension k_{out}, Lemma 3.1 already gives k_{out} actual witnesses and a circuit of size $k_{\text{out}}(S+1)-1$; if the dimension is zero, then $g=0$. The dense simulation below is retained because it isolates transition cost and may matter when coefficient tables are sparse or algorithmically available.

One sufficient certificate uses changing linear spaces. For subspaces $E, F \le \mathcal{R}$, let $E \odot F = \operatorname{span}\{u \odot v : u \in E, v \in F\}$. Assign a subspace E_w to every wire, containing every realized profile, such that NOT includes the constant one and its child space, while an AND output contains the product of its child spaces.

Proposition 6.1 (Profile-space simulation; PROVED HERE)**.** *For a supplied AND/NOT verifier with T gates, put $k_w = \dim E_w$. Over the mixed basis $\{\wedge, \vee, \oplus, \neg\}$, its projection has size at most*

$$\sum_{w=\neg u} k_w(k_u+1) + \sum_{w=u\wedge v} (k_w+1)k_u k_v + \max\{0, k_{\text{out}}-1\}. \tag{6.2}$$

The standard basis changes this by at most a factor four. In particular, if all dimensions are at most k, the bound is $O(Tk^3+k)$. A standard-basis size-S circuit can be converted to an AND/NOT circuit at constant cost, so an $O(Sk^3+k)$ conclusion follows when an admissible dimension-k family is supplied for that converted circuit.

Proof. Choose a basis of each E_w and maintain the coefficient vector of $V_w(x)$. A deterministic input has profile $x_i\mathbf{1}$ and a witness input has a fixed profile, so their coefficients use only input wires and constants. NOT is an affine linear map and costs at most $k_w(k_u+1)$. At an AND gate, expand every product of one child basis vector with another in the output basis. Compute the $k_u k_v$ products of input coefficients once; the k_w output coordinates are fixed parities of these products. This gives the second sum. Finally, the output profile is nonzero precisely when its coefficient vector is nonzero, and the last term is an OR tree. XOR has a four-gate implementation in the standard basis. □

Profile propagation therefore becomes interesting only when the bilinear coefficient tables used by the gate transitions are already available or unusually sparse, when uniform algorithmic access matters, or when one uses nonlinear codes tailored to the realized state pairs.

Multiplication-friendly codes suggest language for the last possibility: structured linear encodings can support reconstruction of products [4]. Existing constructions normally expand a lower-dimensional algebra into enough coordinates for multiplication. They do not compress the full algebra $\mathbb{F}_2^{2^M}$. Any application here must control reachable subspaces or change representations between gates; otherwise Theorem 5.1 applies.

6.2 DNNF and bounded width

There is a clean established regime in which existential projection really is cheap. A decomposable negation normal form (DNNF) is a negation normal form in which the two children of every AND gate mention disjoint variable sets.

Proposition 6.2 (DNNF forgetting; KNOWN). *If a DNNF of edge size T for $f(x, y)$ is supplied, all y variables can be existentially forgotten in linear time without asymptotically increasing size. The resulting DNNF computes g [7, 8].*

The operation replaces every positive or negative literal of a forgotten variable by true and simplifies. Existential quantification distributes over OR. At a decomposable AND, the forgotten variables of the children are disjoint, so their witnesses can be chosen independently and quantification also distributes. Without decomposability, the rule is false: the formula $y \wedge \neg y$ would incorrectly become true.

The phrase "a DNNF is supplied" is essential: the cited compilation theorem does not give a polynomial-size DNNF for an arbitrary circuit. A concrete positive case is a CNF F whose interaction graph has treewidth w; it compiles to DNNF in size and time $2^{O(w)}|F|$ under the source's edge-size convention [7, 8]. If one insists that projection preserve a complete structured deterministic DNNF, width can grow exponentially; Capelli and Mengel give width-parameterized upper and lower bounds [2]. These results make the missing invariant tangible: decomposition or separator width tells the circuit how summaries combine.

Deterministic, smooth DNNF also supports bottom-up model counting and weighted model counting [5]. That is stronger than the existence test needed here. Counting and parity aggregates provide useful algorithmic routes only when the representation makes their sum-product evaluation valid; generic exact counting hardness is context, not an unrestricted circuit lower bound [21].

7 Approximation and the projection metric

An approximant to the verifier need not approximate its existential projection. This matters because approximation in the companion manuscript is measured on the full (x, y) cube.

Proposition 7.1 (Projection can amplify error; PROVED HERE). *Let f' differ from f on an ε fraction of uniformly random pairs (x, y), and let $g'(x) = \bigvee_y f'(x, y)$. Then*

$$\Pr_x[g(x) \neq g'(x)] \leq \min\{1, 2^M \varepsilon\}, \tag{7.1}$$

and the factor 2^M is tight.

Proof. Every x on which the projections differ contributes at least one erroneous pair (x, y). Divide the number of erroneous pairs by 2^N. For tightness, take $f(x, y) = 1[y = 0^M]$ and $f' = 0$. Their distance is 2^{-M}, while their projections are the constants one and zero. ☐

There are two correct one-sided alternatives.

- The parity fingerprint in Theorem 4.2 controls error directly over x, independently of fiber density, but still needs an aggregate evaluator.
- If nonempty fibers have density δ, sample t actual witnesses and OR their verifier values. A fixed yes-input is missed with probability at most $(1-\delta)^t$, there are no false positives, and the circuit uses at most $tS + t - 1$ gates. Averaging fixes one sample with the same error bound under any chosen distribution on x.

Uniform (x, y) approximation, direct approximation of the projection, and a per-fiber density guarantee are therefore different notions. Only the last two control the decision problem in (1.1) without the 2^M amplification.

8 Circuit-complexity consequences

It is useful to state the parameter comparisons without complexity-class shorthand. If $K = 0$, then $g = 0$. Otherwise, let B_{agg} denote the cost of a supplied joint circuit for the r parity aggregates. Up to lower-order and fixed-basis factors, the available envelope is

$$\mathrm{C}(g) \le \min\left\{2^M(S+1), \frac{2^N}{N}, R(S+1), \frac{\log(K+1)}{\delta}(S+1), B(M,d)(S+1), B_{\text{agg}} + r\right\}, \tag{8.1}$$

where each structural term is present only under its stated hypothesis. The imported mass-production term is valid in the fixed-ratio regime but is omitted from (8.1) because direct synthesis generically dominates it.

The replication term beats direct synthesis when roughly

$$M + \log_2(S+1) < N - \log_2 N. \tag{8.2}$$

The dense-fiber term is a genuine improvement only if its number of retained witnesses is below both 2^M and $2^N/(N(S+1))$. Similar comparisons apply to R and $B(M, d)$. A reduction in witnesses is not automatically a reduction below the best circuit already available for g.

8.1 Why a universal polynomial determinizer would be major

Suppose there were a constant c such that every relation with $M \le \mathrm{poly}(N)$ satisfied

$$\mathrm{C}(\exists y\, f) \le (N + M + \mathrm{C}(f) + 1)^c. \tag{8.3}$$

Here NP/poly is the class recognized by polynomial-size nonuniform verifier circuits with polynomially many witness bits. Applying the hypothesis to those verifiers would give NP/poly $\subseteq$ P/poly; the reverse inclusion is trivial, so equality follows, and in particular NP $\subseteq$ P/poly. The Karp–Lipton theorem would then collapse the polynomial hierarchy to its second level [13]. A stronger uniform compiler that takes a polynomial-size verifier description, runs in polynomial time, and outputs a polynomial-size circuit for its projection would imply P = NP. These are conditional consequences, not results of this note.

The aggregate barrier has parallel context. For a uniform verifier family, $\bigoplus_y f(x, y)$ is a parity-counting predicate, while the integer sum is a #P computation. For polynomial-size nonuniform verifier families, a polynomial circuit bound for every parity aggregate would give $\oplus\mathrm{P/poly} = \mathrm{P/poly}$. Uniform counting hardness does not itself prove a Boolean circuit lower bound, so the rigorous unconditional obstruction remains Theorem 5.1.

Unambiguity does not make the aggregate free. If every fiber has size at most one, OR equals parity exactly, but evaluating that parity may still be hard. Valiant–Vazirani's isolation theorem reinforces the distinction between obtaining a unique-witness promise and deterministically evaluating the resulting language [22].

For comparison, easy-witness results compress the *description of a witness* under circuit assumptions; they do not supply one circuit computing the projection for every x [17]. Work on nondeterministic circuit lower bounds also addresses a different direction. For example, Morizumi gives restricted-basis lower bounds and an explicit Tseitin-style translation from circuits to nondeterministic 3-CNFs [15]. None of those results turns the measurement reduction in Theorem 4.1 into a generic gate bound.

There is, however, an elementary lower-bound transfer. If $g = \exists_y f$ and $\mathrm{C}(g) \geq L$, then (1.2) implies

$$\mathrm{C}(f) \geq \frac{L+1}{2^M} - 1. \tag{8.4}$$

The factor 2^M makes this weak when the witness budget is large.

9 The resulting research program

The mass-production viewpoint does inspire a technical target for this approach, but not a finished generic determinization theorem. It says to regard all branches with the same deterministic input as one coded object and to share the work that produces summaries of that object. The preceding results narrow the target to four forms.

1. **Boundary-efficient coded recovery.** Choose a zero-preserving fingerprint together with recovery representations whose symmetric-difference boundary Z is small, as in Proposition 4.4, and prove that the surviving resource map G_Z has a cheap joint circuit. The objective is the full right side of (4.9), not just the number of fingerprint bits.
2. **Changing multiplicative fingerprints.** Use different encodings at different wires and require transitions only on profiles reached by the verifier. The fixed-sketch ideal theorem no longer applies, but every transition must have an explicit small circuit. Restricted product spaces, sparse bilinear transition tables, and multiplication-friendly codes are plausible tools.
3. **Succinct actual-witness hitting.** The dense and low-degree arguments give small hitting sets, but the relation-dependent choices inspect the fibers. Find checkable promises under which a small H can be constructed from a succinct verifier without enumerating its truth table.
4. **Restricted-model lower bounds.** Exhibit verifier families for which every zero-preserving aggregate system has large transition cost, even though its output width is small. The ideal-kernel theorem resolves only the fixed linear/full-algebra model; lower bounds for changing or nonlinear sketches are not resolved here.

The central compression hierarchy is therefore

$$\boxed{\begin{array}{c}\text{fiber-code dimension or density from zero} \Rightarrow \text{few actual witnesses} \Rightarrow \text{a circuit bound,}\\ \text{few realized fibers} \Rightarrow \text{few parity measurements} \not\Rightarrow \text{a circuit bound,}\\ \text{small recovery boundary plus cheap shared resources} \Rightarrow \text{a circuit bound.}\end{array}} \tag{9.1}$$

The second implication becomes useful precisely when a structural representation supplies the missing aggregate computation. That is the classical nondeterministic-sharing problem left by this note.

Reproducibility note

The accompanying research corpus contains an exhaustive finite checker for small instances of Theorems 4.1 and 5.1 and Lemmas 3.1, 3.4, 3.5 and 4.3, including affine-line cancellation and the two-bit AND collision. These checks are evidence against transcription and edge-case errors; they are not substitutes for the proofs above. The source registry separately records the primary-source boundary for every imported or known result.

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